

gate algo

Lec-54

Insertion in heap

Heapify → Insertion  
in heap

Lec\_55

## Deletion in Heap

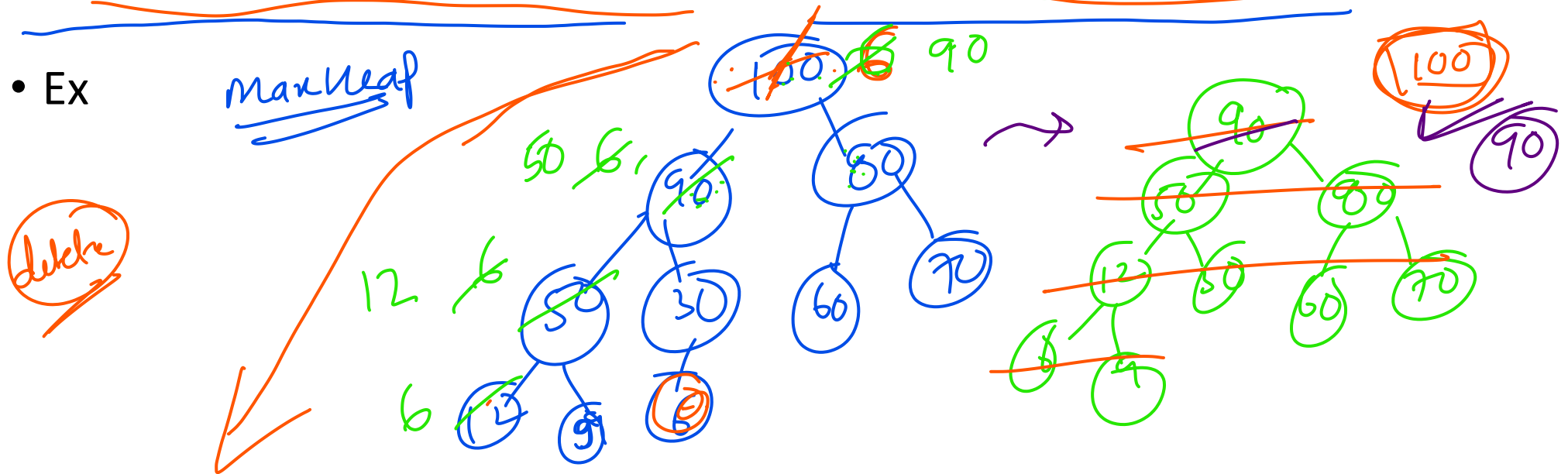
Insertion:

always Insertion happens at the first

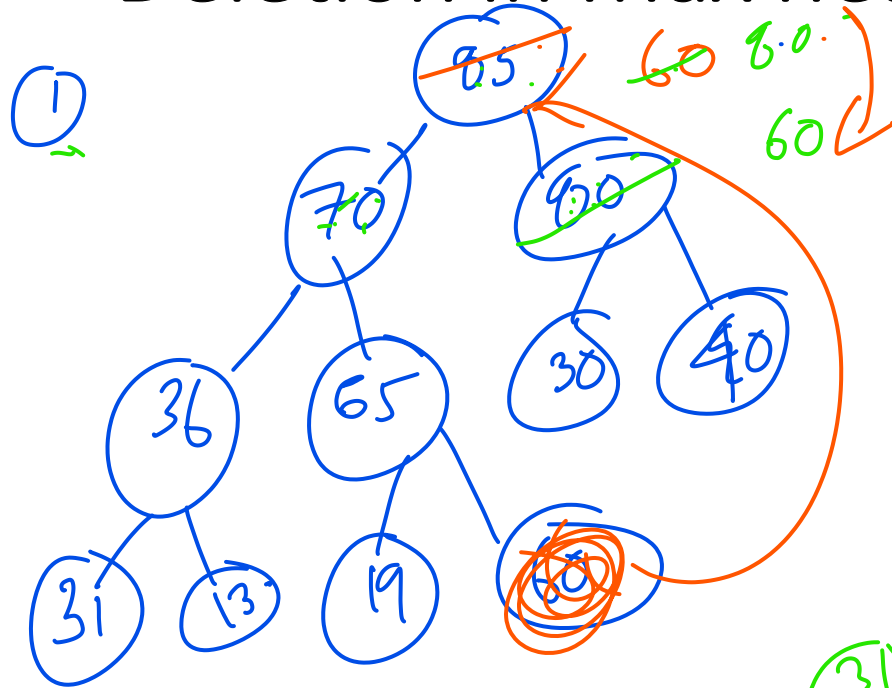
empty slot / then modify

In any heap always root element is deleted and replaced by the last element then make the changes

• Ex

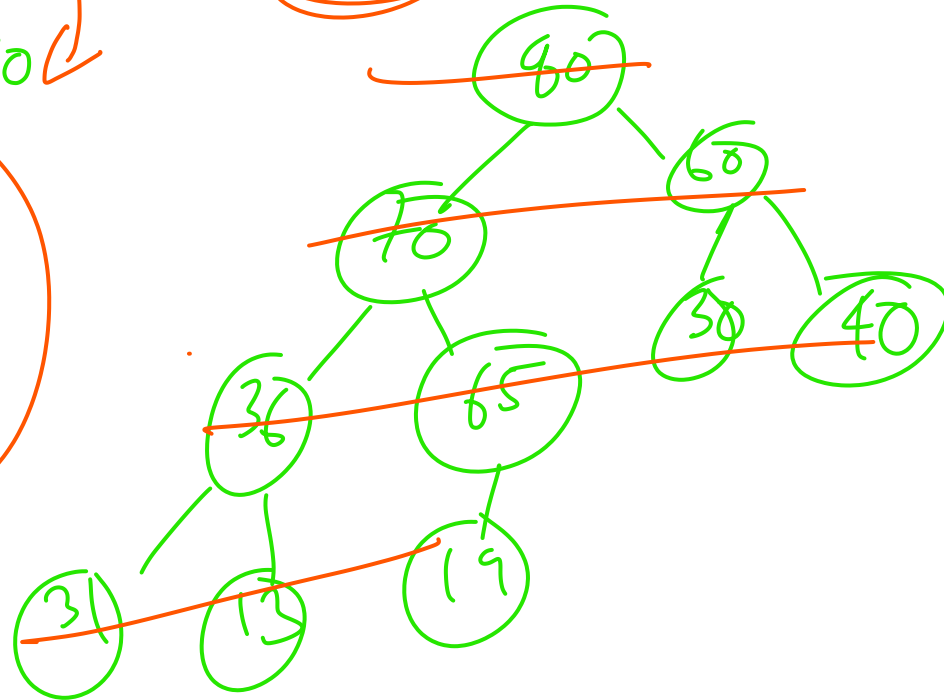


# Deletion in Max heap:



delete <sup>del</sup> root element

95



→ delete root  $\Rightarrow O(1)$

→ replace by last elem  $\Rightarrow O(1)$

→ for going to BC  $\Rightarrow O(1)$   
WC  $\Rightarrow O(\log n)$

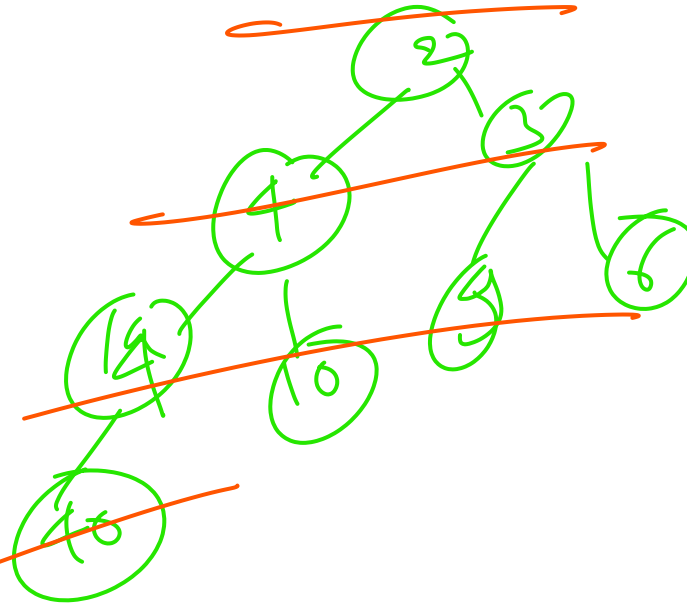
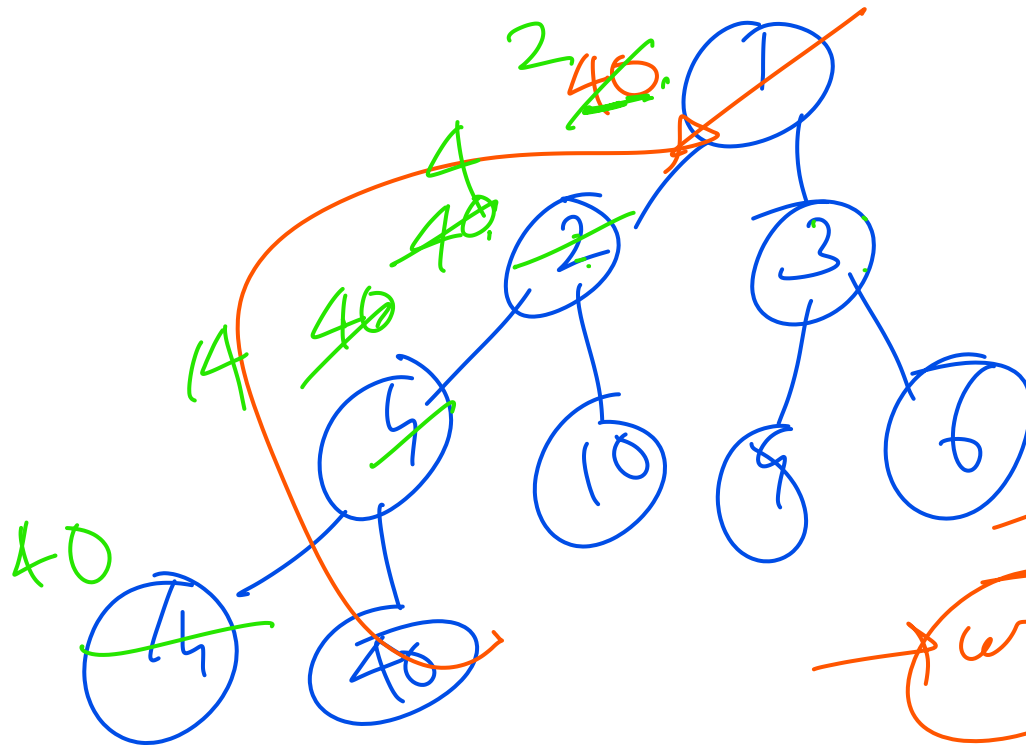
For deletion  $\Rightarrow$

|                  |                            |
|------------------|----------------------------|
| TC $\Rightarrow$ | BC $\Rightarrow O(1)$      |
| Ac)              | WC $\Rightarrow O(\log n)$ |



# Deletion in Min Heap:

delete  $\Rightarrow$  deletion of root



WC

$$T_C \Rightarrow BC = O(1)$$

$$AC, WC = O(\log n)$$





## Time and Space Complexity:

→ for both the heap

In BC,  $TC = O(1)$

$WC = \underline{O(\log n)}$

→ ★ For Insertion or deletion or heapify

$TC \Rightarrow BC \Rightarrow O(1)$

$WC, AC \Rightarrow O(\log n)$



## Important Points:

max1, max2, max3, ...

- In max heap always root element is deleted so in first deletion first max and in second deletion second max will come
- In min heap always root element is deleted so in first deletion first min and in second deletion second min will come
- In n node max-heap or min heap tree to delete or insert 1 element

WC and AC Time Complexity:  $O(\log n)$

BC:  $O(1)$

And this method is called heapify method

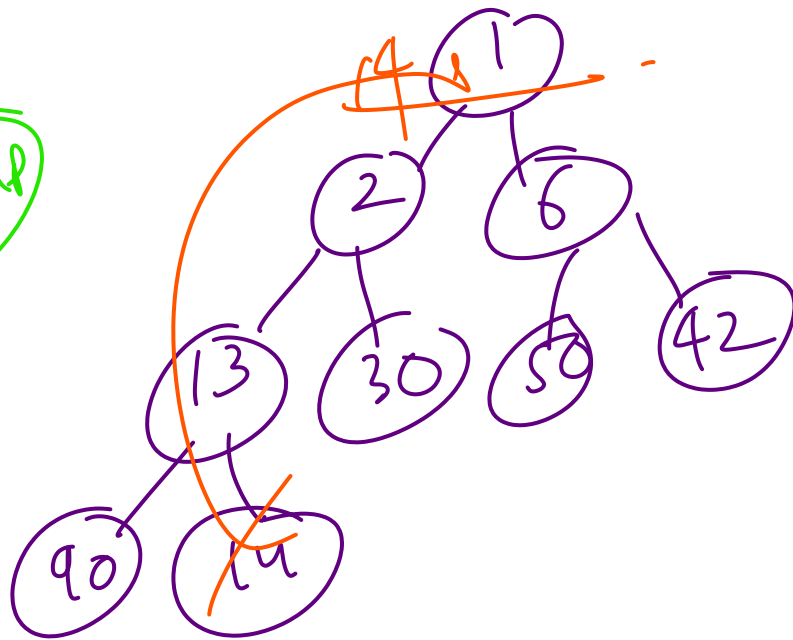
In-place and Not stable

Heap  
selection  
quick

Outplace = merge sort

#

min heap



|   |   |   |    |    |    |    |    |    |
|---|---|---|----|----|----|----|----|----|
| 1 | 2 | 6 | 13 | 30 | 50 | 42 | 90 | 14 |
|---|---|---|----|----|----|----|----|----|

|   |   |   |    |    |  |  |  |  |
|---|---|---|----|----|--|--|--|--|
| 1 | 2 | 6 | 13 | 42 |  |  |  |  |
|---|---|---|----|----|--|--|--|--|

Sorted